VQE for the Hubbard ring on MIMIQ: the equations, step by step

Khuram Shahzad, October 2026

The ground state of the Hubbard ring is prepared with the variational quantum eigensolver (VQE) on MIMIQ and passed on to the nBRDM and GME analysis. All the equations needed for this are given below, step by step. Sites k and orbitals p are counted from 1, qubits from 0. Every formula has been checked numerically (Section 10).

Symbols. t, U, V: model parameters; $\mathcal{U}(\boldsymbol{\theta})$: the UCCSD circuit; $R_P(\varphi)$: the Pauli rotation gate; $\mathcal{R}_C$: the rotation from the orbitals to the sites; q: a qubit index; $c_{k,\sigma}$ ($a_{p,\sigma}$): annihilation operator on site k (orbital p) with spin σ.

The procedure in short

One Hamiltonian. The Hamiltonian is always the full H_e of Eq. (1), with t, U and V. It is never changed, and VQE never diagonalizes it: VQE only measures its expectation value. (The exact diagonalization, ED, of H_e is used only as the reference to compare with.)

Two bases for the circuit. The qubits can stand for two different sets of one-electron states: the *sites* (an electron on site k) or the *orbitals* (an electron in a wave spread over the ring). Both describe the same space; a change of basis changes how states look, not the physics. The orbitals are used only to *build the starting state*, because there it is a few X gates; the energy and the nBRDMs are always measured on the sites.

The steps, in order.

1. Hamiltonian $H_e(t, U, V)$ on the sites (Section 1).
2. One qubit per spin orbital and the Jordan–Wigner transformation: H_e becomes a sum of Pauli strings on the sites (Section 2).
3. Orbitals from the $L \times L$ one-electron hopping matrix, only to prepare the state (Section 3).
4. Starting state: Hartree–Fock, the lowest orbitals filled, i.e. X gates (Section 4).
5. Ansatz: UCCSD with parameters $\boldsymbol{\theta}$, acting on the orbitals (Section 5).
6. Rotation of the state from the orbitals to the sites (Section 6).
7. Energy $E(\boldsymbol{\theta}) = \langle H_e \rangle$ measured on MIMIQ, with U and V included (Section 7).
8. COBYLA changes $\boldsymbol{\theta}$ until $E(\boldsymbol{\theta})$ stops decreasing (Section 8).
9. From the final state: nBRDMs and GME, compared with ED (Section 9).

The circuit is therefore

$$|0\cdots0\rangle \xrightarrow{X \text{ gates}} |\Phi_0\rangle \xrightarrow{\mathcal{U}(\boldsymbol{\theta})} |\Psi_{\text{orb}}(\boldsymbol{\theta})\rangle \xrightarrow{\mathcal{R}_C} |\Psi_e(\boldsymbol{\theta})\rangle \;\rightarrow\; \text{measure } \langle H_e \rangle,$$

where the first three objects are written in the orbitals and the last one on the sites.

1 The Hubbard Hamiltonian

The one-dimensional extended Hubbard model with K electrons on a ring of L sites is defined by the Hamiltonian

$$H_e = -t \sum_{k=1}^{L} \sum_{\sigma=\uparrow,\downarrow} \left(c^\dagger_{k,\sigma}\, c_{k+1,\sigma} + c^\dagger_{k+1,\sigma}\, c_{k,\sigma} \right) + U \sum_{k=1}^{L} n_{k,\uparrow}\, n_{k,\downarrow} + V \sum_{k=1}^{L} n_k\, n_{k+1} + \epsilon \sum_{k=1}^{L} n_k \,. \tag{1}$$

Here $n_{k,\sigma} = c^\dagger_{k,\sigma} c_{k,\sigma}$ is the number operator of site k ($1 \le k \le L$) with spin $\sigma = \uparrow, \downarrow$, and $n_k = n_{k,\uparrow} + n_{k,\downarrow}$ the number of electrons on site k; t, U and V are the hopping, on-site and nearest-neighbor interaction parameters. The sum over sites is written out for each term: the hopping has two sums, over the bonds $(k, k+1)$ and over the spin, since an electron of either spin can hop and the hopping does not flip the spin. Because of the periodic boundary conditions, site $L+1$ coincides with site 1. The detection orbital ϕ_k is the one localized on site k, and the spins are extracted from n consecutive orbitals $\phi_1, \ldots, \phi_n$. At fixed K the ϵ term is the constant ϵK; we use $\epsilon = 0$. We work at half filling, $K = L$, in the sector with $K_\uparrow = K_\downarrow = K/2$ electrons of each spin.

Taking t out of every term gives $H_e(t, U, V) = t\, H_e(1, U/t, V/t)$. Multiplying by $t > 0$ rescales all energies by t but leaves the eigenstates unchanged. Hence the ground state $|\Psi_0\rangle$ depends only on U/t and V/t, and its energy is $E_0(t, U, V) = t\, E_0(1, U/t, V/t)$.

2 Spin orbitals, qubits and Jordan–Wigner

Each spin orbital (k, σ) is stored on one qubit; qubits are counted from 0:

$$q(k, \sigma) = 2(k-1) + \sigma, \qquad \sigma = 0\ (\uparrow),\ 1\ (\downarrow), \tag{2}$$

so that site k is stored on the two qubits with labels $2k-2$ ($\uparrow$) and $2k-1$ ($\downarrow$). The labels run from 0 to $2L-1$, so the ring needs $2L$ qubits (12 for $L = 6$, labels $0, \ldots, 11$). A qubit in $|1\rangle$ means the spin orbital is occupied. The Jordan–Wigner transformation writes the operators of Eq. (1) with Pauli matrices on these qubits. With $q = q(k, \sigma)$,

$$c_{k,\sigma} = Z_0 Z_1 \cdots Z_{q-1}\, \frac{X_q + i Y_q}{2}, \qquad n_{k,\sigma} = c^\dagger_{k,\sigma} c_{k,\sigma} = \frac{1 - Z_q}{2}. \tag{3}$$

Here $c_{k,\sigma}$ removes the electron from the spin orbital (k, σ) (it maps $|1\rangle_q \to |0\rangle_q$), and $n_{k,\sigma}$ is its occupation, 0 (empty, $Z_q = +1$) or 1 (occupied, $Z_q = -1$); the string $Z_0 \cdots Z_{q-1}$ gives the fermion sign. For the three types of term in Eq. (1) this gives, with $q = q(k, \sigma)$ and $q' = q(k+1, \sigma)$ in the hopping, ordered so that $q < q'$,

$$c^\dagger_{k,\sigma} c_{k+1,\sigma} + c^\dagger_{k+1,\sigma} c_{k,\sigma} = \tfrac{1}{2} \left(X_q Z_{q+1} \cdots Z_{q'-1} X_{q'} + Y_q Z_{q+1} \cdots Z_{q'-1} Y_{q'} \right), \tag{4}$$

$$n_{k,\sigma}\, n_{k',\sigma'} = \tfrac{1}{4} \left(1 - Z_q - Z_{q'} + Z_q Z_{q'} \right), \qquad q = q(k, \sigma) \neq q' = q(k', \sigma'), \tag{5}$$

where Eq. (5) gives the U term with $k' = k$, $\sigma = \uparrow$, $\sigma' = \downarrow$, and the V term through $n_k n_{k+1} = \sum_{\sigma,\sigma'} n_{k,\sigma}\, n_{k+1,\sigma'}$. In Eq. (4), for $k < L$, $q' = q + 2$ and the string is the single Z_{q+1} of the other spin on site k; for the ring bond ($k = L$, $k + 1 \equiv 1$) the two qubits are σ and $2L - 2 + \sigma$, and the string covers all the qubits in between. Example: hopping of $\uparrow$ between sites 1 and 2 is $\frac{1}{2}(X_0 Z_1 X_2 + Y_0 Z_1 Y_2)$. The Hamiltonian therefore becomes a sum of Pauli strings with real coefficients,

$$H_e = \lambda_0 + \sum_{\mu} \lambda_\mu P_\mu, \qquad P_\mu \in \{I, X, Y, Z\}^{\otimes 2L}, \tag{6}$$

with the constant $\lambda_0 = UL/4 + VL$ at $\epsilon = 0$. For $L = 6$ there are 42 strings at $(t, U, V) = (1, 4, 0)$ (24 hopping, 18 on-site) and 66 at $(1, 10, 5)$.

Bit order: MIMIQ takes qubit 0 as the lowest bit of a basis state. This matters only when a state vector is loaded or read out and compared with another convention; gates and Pauli strings refer to qubit indices and do not depend on it.

3 Orbitals for preparing the state

This section does not change the Hamiltonian: H_e stays the full Hamiltonian of Eq. (1), with U and V. What is needed here is a starting state for the circuit that is easy to prepare with gates and already close to the ground state. It is built from one-electron states (orbitals) obtained from the hopping term alone, for three reasons:

- the hopping is the only one-electron part of H_e, so it defines one-electron states; U and V are interactions between electrons, and they are already diagonal on the sites (they only count electrons);
- at $U = V = 0$, H_e is the hopping alone, and its exact ground state is obtained by filling the lowest orbitals (Section 4); for $U \neq 0$ this state is the starting point that VQE improves;
- finding the orbitals needs only the diagonalization of a small $L \times L$ matrix, whereas the exact ground state needs the diagonalization of H_e in the full many-electron space (924 states for $L = 6$, growing exponentially with L), which VQE avoids.

The hopping matrix of the ring (one electron, $L \times L$) is

$$h_{kk'} = -t\left(\delta_{k',k+1} + \delta_{k',k-1}\right), \qquad 1 \le k, k' \le L \ \ (\text{site } L+1 \equiv 1), \tag{7}$$

and its eigenvectors define the orbitals $p = 1, \ldots, L$, sorted by energy,

$$h = C\,\mathrm{diag}(e_1, \ldots, e_L)\,C^T, \qquad c_{k,\sigma} = \sum_{p=1}^{L} C_{kp}\, a_{p,\sigma}, \tag{8}$$

with C real and orthogonal (row k: site, column p: orbital). The columns of C are the eigenvectors of h; they form the new basis (the orbitals), and the eigenvalues e_p are the orbital energies. For $L = 6$: $e_p = -2t, -t, -t, t, t, 2t$; the lowest orbital is spread evenly over the ring, $C_{k1} = 1/\sqrt{6}$, and the higher ones change sign more often from site to site. The orbitals and the sites are two bases of the same one-electron space, related by the orthogonal matrix C; nothing is lost in the change. The orbitals depend only on t, not on U or V; they play the role of the Hartree–Fock orbitals of a molecule. In the orbital basis the spin orbital (p, σ) is stored on qubit $q(p, \sigma) = 2(p-1) + \sigma$, the same rule as Eq. (2).

In the orbital basis the hopping is diagonal but the U and V terms are not: writing H_e in the orbitals (substituting Eq. (8) into Eq. (1)) gives the same spectrum but many more Pauli strings (306 instead of 42 at $(1, 4, 0)$). On the sites it is the other way round. No basis makes both parts diagonal; this is why the problem is hard. The state is therefore prepared in the orbitals and rotated back to the sites (Section 6) before the full H_e is measured.

4 Hartree–Fock reference

The reference state fills the $K/2$ lowest orbitals with both spins,

$$|\Phi_0\rangle = \prod_{p=1}^{K/2} a^\dagger_{p,\uparrow}\, a^\dagger_{p,\downarrow}\, |\mathrm{vac}\rangle, \tag{9}$$

that is, the qubits $0, \dots, K-1$ are set to $|1\rangle$ by X gates. Since the density is uniform ($\langle n_{k,\sigma}\rangle = 1/2$) and the on-site term has no exchange part,

$$E_{\mathrm{HF}} = 2\sum_{p=1}^{K/2} e_p + U\sum_{k=1}^{L}\langle n_{k,\uparrow}\rangle\langle n_{k,\downarrow}\rangle = 2\sum_{p=1}^{K/2} e_p + \frac{UL}{4} \overset{L=6}{=} -8t + \tfrac{3}{2}U \qquad (\epsilon = V = 0). \tag{10}$$

$|\Phi_0\rangle$ is the exact ground state of H_e only at $U = V = 0$; for $U > 0$ it is the starting point of VQE, and Eq. (10) is its energy with the full H_e. The reference must be a closed shell: the last filled level must not be degenerate with the first empty one. This holds for $L = 6$ (levels $-2t, -t, -t \,|\, t, t, 2t$) but not for $L = 4$ ($-2t, 0 \,|\, 0, 2t$). At $U/t = 1, 4, 10$, $|\Phi_0\rangle$ (rotated to the sites) has the fidelity 0.980, 0.689 and 0.265 with the exact ground state, against 3–19 % for the best single configuration on sites.

5 The ansatz: UCCSD in the orbital basis

Let i, j run over the occupied orbitals $(1, \dots, K/2)$ and a, b over the empty ones $(K/2{+}1, \dots, L)$. The cluster operator contains all the spin-conserving single and double excitations,

$$T(\boldsymbol{\theta}) = \sum_{\sigma}\sum_{a,i} \theta^{\sigma}_{ai}\, a^{\dagger}_{a,\sigma} a_{i,\sigma} + \sum_{\text{doubles}} \theta_{a\sigma\, i\sigma\, b\sigma'\, j\sigma'}\, a^{\dagger}_{a,\sigma} a_{i,\sigma}\, a^{\dagger}_{b,\sigma'} a_{j,\sigma'}, \tag{11}$$

where the doubles are of three kinds: $\uparrow\downarrow$ (all a, i, b, j), $\uparrow\uparrow$ and $\downarrow\downarrow$ ($a < b$, $i < j$). With $o = K/2$ occupied and $v = L - K/2$ empty orbitals the number of parameters is

$$N_\theta = \underbrace{2ov}_{\text{singles}} + \underbrace{(ov)^2}_{\uparrow\downarrow} + \underbrace{2\tbinom{o}{2}\tbinom{v}{2}}_{\uparrow\uparrow,\ \downarrow\downarrow} \overset{o=v=3}{=} 18 + 81 + 18 = 117. \tag{12}$$

The parameters are ordered as singles $\uparrow$, singles $\downarrow$, doubles $\uparrow\downarrow$, $\uparrow\uparrow$, $\downarrow\downarrow$. The UCCSD state is

$$|\Psi_{\mathrm{orb}}(\boldsymbol{\theta})\rangle = e^{\,T(\boldsymbol{\theta}) - T(\boldsymbol{\theta})^{\dagger}} |\Phi_0\rangle. \tag{13}$$

Since $T - T^{\dagger}$ is anti-Hermitian, $G = i(T - T^{\dagger})$ is Hermitian, and Jordan–Wigner turns it into Pauli strings with real coefficients that are linear in $\boldsymbol{\theta}$:

$$e^{T - T^{\dagger}} = e^{-iG}, \qquad G(\boldsymbol{\theta}) = \sum_{m} \gamma_m(\boldsymbol{\theta})\, Q_m. \tag{14}$$

On MIMIQ the exponential is applied with one first-order Trotter step,

$$\mathcal{U}(\boldsymbol{\theta}) = \prod_{m} e^{-i\gamma_m Q_m} = \prod_{m} R_{Q_m}(2\gamma_m), \qquad R_P(\varphi) = e^{-i\varphi P/2}, \tag{15}$$

where R_P is the Pauli rotation gate. The strings Q_m do not all commute, so Eq. (15) is not exactly Eq. (13); it *defines* the ansatz, and the optimizer adjusts $\boldsymbol{\theta}$ for it. Every excitation conserves the number of electrons and S_z, so every factor of Eq. (15) does too; the total spin S^2 is not conserved, because the 117 amplitudes are independent (not spin adapted). The circuit starts with the X gates of Eq. (9). After decomposition into one- and two-qubit gates the circuit has 29 298 gates (9 024 CNOTs) for $L = 6$.

6 Rotation from the orbitals back to the sites

$|\Psi_{\mathrm{orb}}\rangle$ is written in the orbital modes $a_{p,\sigma}$. We need the same physical state written in the site modes $c_{k,\sigma}$, on the same qubits. This is done by the unitary $\mathcal{R}_C$ defined by

$$\mathcal{R}_C\, a^{\dagger}_{p,\sigma}\, \mathcal{R}_C^{\dagger} = \sum_{k=1}^{L} C_{kp}\, c^{\dagger}_{k,\sigma}, \qquad \mathcal{R}_C|\mathrm{vac}\rangle = |\mathrm{vac}\rangle, \tag{16}$$

which replaces every orbital creation operator by the combination of site operators of Eq. (8). The state on the sites is then

$$|\Psi_e(\boldsymbol{\theta})\rangle = \mathcal{R}_C\,\mathcal{U}(\boldsymbol{\theta})\,X_0X_1\cdots X_{K-1}|0\cdots0\rangle. \tag{17}$$

Givens decomposition. The orthogonal matrix C is brought to a diagonal matrix of signs by rotations W_r of two neighboring rows $(p, p+1)$,

$$W_{N_g}\cdots W_2\,W_1\,C = D, \qquad D = \mathrm{diag}(\pm1), \qquad N_g = \tfrac{L(L-1)}{2} = 15, \tag{18}$$

each rotation zeroing one element below the diagonal. A rotation by ϑ of the orbitals p and $p+1$ is, for both spins, the fermionic unitary

$$g_p(\vartheta) = \exp\Big[\vartheta\sum_\sigma\big(a^\dagger_{p+1,\sigma}a_{p,\sigma} - a^\dagger_{p,\sigma}a_{p+1,\sigma}\big)\Big]. \tag{19}$$

With $q = q(p,\sigma)$, the two orbitals sit on the qubits q and $q+2$, with the other spin on $q+1$ in between, and Jordan–Wigner gives exactly two commuting strings,

$$i\vartheta\big(a^\dagger_{p+1,\sigma}a_{p,\sigma} - a^\dagger_{p,\sigma}a_{p+1,\sigma}\big) = \frac{\vartheta}{2}\big(X_qZ_{q+1}Y_{q+2} - Y_qZ_{q+1}X_{q+2}\big), \tag{20}$$

so that, without any Trotter error,

$$g_p(\vartheta) = \prod_\sigma R_{X_qZ_{q+1}Y_{q+2}}(\vartheta)\,R_{Y_qZ_{q+1}X_{q+2}}(-\vartheta). \tag{21}$$

The whole rotation $\mathcal{R}_C$ is a Z gate on the qubits $q(p,\uparrow)$ and $q(p,\downarrow)$ for every $D_{pp} = -1$, followed by the N_g rotations of Eq. (18) in reverse order: $4N_g = 60$ Pauli rotations for $L = 6$. Which signs occur depends on the numerical eigenvectors: at $t = 1$ all $D_{pp} = +1$, at $t = 0.05$ $D_{66} = -1$.

Why rotate instead of measuring in the orbital basis: the energy then needs only the site strings of Eq. (6) (42 instead of 306), the nBRDMs are defined on sites and can be measured directly, and the Hamiltonian is used unchanged.

7 The energy measured on MIMIQ

For given $\boldsymbol{\theta}$ the circuit of Eq. (17) is run on MIMIQ, and the only output is the expectation value

$$E(\boldsymbol{\theta}) = \langle\Psi_e(\boldsymbol{\theta})|H_e|\Psi_e(\boldsymbol{\theta})\rangle = \lambda_0 + \sum_\mu \lambda_\mu\langle P_\mu\rangle_{\boldsymbol{\theta}}, \tag{22}$$

obtained with one expectation-value instruction for the Hamiltonian of Eq. (6), the constant λ_0 added separately. On the exact simulator (Exaqt) $\langle P_\mu\rangle$ is exact; on the digital twin or a device each $\langle P_\mu\rangle$ is estimated from a finite number of shots. By the variational principle $E(\boldsymbol{\theta}) \geq E_0$ for every $\boldsymbol{\theta}$.

8 The optimization

The parameters are found by minimizing Eq. (22),

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} E(\boldsymbol{\theta}), \qquad E_{\mathrm{VQE}} = E(\boldsymbol{\theta}^*), \tag{23}$$

with COBYLA, a derivative-free optimizer that needs only the energy values:

- start: $\boldsymbol{\theta} = 0$, that is $|\Phi_0\rangle$ and $E = E_{\mathrm{HF}}$;
- chunks of 600 COBYLA iterations ($\rho_{\mathrm{beg}} = 0.2$, tolerance 10^{-10}); each chunk restarts from the best point so far, moved by a random step of size 5×10^{-4};
- stop when the best energy has improved by less than 10^{-6} in 3 consecutive chunks, or after at most 25 chunks (15 000 energy evaluations).

CCSD amplitudes are not used as a start: with $t = 1$ CCSD gives $E_{\mathrm{CCSD}} = -6.6011$, -3.7171 and -5.5382 at $U = 1$, 4 and 10, against $E_0 = -6.6012$, -3.6687 and -1.6644; at large U it is far below E_0 (it is not variational).

9 After convergence: nBRDMs and GME

The state $|\Psi_e(\boldsymbol{\theta}^*)\rangle$ is prepared again with the same circuit, and the nBRDM of each cluster of n consecutive sites is measured (orbitals $\phi_1, \ldots, \phi_n$ and their translations) from Pauli strings, exactly as for the exact state:

$$\Gamma^{(n)}_{\boldsymbol{\alpha},\boldsymbol{\alpha}'} = \sum_{\nu} w_\nu^{\boldsymbol{\alpha}\boldsymbol{\alpha}'} \langle P_\nu \rangle_{\boldsymbol{\theta}^*}, \qquad \rho_s = \frac{\Gamma^{(n)}}{\mathrm{Tr}\,\Gamma^{(n)}}, \tag{24}$$

where $\boldsymbol{\alpha}, \boldsymbol{\alpha}'$ label the spin configurations of the n sites, P_ν are the Pauli strings on the cluster's $2n$ qubits and w_ν their weights. $\mathcal{C}_{\mathrm{GME}}(\rho_s)$ is computed with the GCG convex-roof solver. Because the VQE state may break the translation symmetry slightly, all L clusters are measured for each n. For every parameter point we report side by side

$$\Delta E = E_{\mathrm{VQE}} - E_0, \quad F = |\langle \Psi_0 | \Psi_e(\boldsymbol{\theta}^*) \rangle|^2, \quad \max |\Gamma^{(n)}_{\mathrm{VQE}} - \Gamma^{(n)}_{\mathrm{ED}}|, \quad |\mathcal{C}^{\mathrm{VQE}}_{\mathrm{GME}} - \mathcal{C}^{\mathrm{ED}}_{\mathrm{GME}}|, \tag{25}$$

where the fidelity F uses the state vector and is available only on the simulator. The parameter points are the U-scan ($t = 1$, $V = 0$, $U = 1, \ldots, 20$), the t-scan ($U = 10$, $V = 0$, $t = 0.05, \ldots, 2$) and the V-scan ($t = 1$, $U = 10$, $V = 0, \ldots, 12$), 65 points in total, each compared with exact diagonalization (ED).

10 What has been checked

Converged UCCSD runs on MIMIQ. $L = 6$, $t = 1$, $V = 0$, the full circuit of Eq. (17) on Exaqt, COBYLA from $\boldsymbol{\theta} = 0$. They agree with UCCSD with the exact exponential of Eq. (13) (no Trotter step, converged with BFGS):

		exact exponential		Trotter step, MIMIQ	
U ($t = 1$)	E_0	ΔE	F	ΔE	F
1	-6.6012	1.0×10^{-4}	0.99999	1.6×10^{-4}	0.99997
4	-3.6687	3.0×10^{-2}	0.9937	–	–
10	-1.6644	3.9×10^{-1}	0.943	3.9×10^{-1}	0.942

UCCSD is accurate at small U/t and degrades at large U/t, where the ground state is far from a single configuration. For large U/t and for the digital twin (the UCCSD circuit is too deep), shallower ansätze such as the Hamiltonian variational ansatz (HVA) and the local unitary cluster Jastrow ansatz (LUCJ) are better suited.

11 Practical points on MIMIQ

- The exact simulator (Exaqt) does not execute Pauli rotation gates directly, also when they sit inside the Trotter block of Eq. (15); the circuit is first decomposed into standard one-

Step	Check	Result
H_e, Eq. (1)	equal to the formula written term by term; Hermitian; conserves K, S_z, S^2; translation invariant	exact (5 points)
H_e on MIMIQ, Eq. (6)	$\langle H_e \rangle$ in the exact ground state $= E_0$	3×10^{-15} ($L = 4, 6$)
Eqs. (4), (5)	coefficients one by one; $\langle H_e \rangle$ on a random circuit	10^{-12}, 10^{-10}
Orbitals, Eq. (8)	E_0 equal in both bases	10^{-10} (5 points)
Hartree–Fock, Eq. (10)	$E(\boldsymbol{\theta} = 0) = -8t + \frac{3}{2}U$ on MIMIQ	6.6×10^{-14}
Rotation, Eq. (16)	exact orbital ground state $\to$ site ground state; one electron in orbital $p \to$ amplitudes C_{kp}	$F = 1.000000000000$ (5 points); 10^{-10}
Eq. (20)	Jordan–Wigner of one Givens generator	two strings, $\pm\vartheta/2$
N_θ, Eq. (12)	excitations counted one by one	$18 + 99 = 117$
UCCSD, Eq. (15)	conserves K, S_z; MIMIQ energy $=$ state-vector energy $\geq E_0$	10^{-10}
nBRDMs	from Pauli strings on MIMIQ vs. ED	10^{-15} ($n \leq 4$)

Table 1: Checks of the steps above. The five points are $(t, U, V) = (1, 1, 0)$, $(1, 4, 0)$, $(1, 10, 0)$, $(0.05, 10, 0)$, $(1, 10, 5)$.

and two-qubit gates.

- A Pauli rotation needs a real angle; coefficients that come out of the Jordan–Wigner transformation as complex numbers with zero imaginary part are passed as their real part.
- The sign step of $\mathcal{R}_C$ applies Z only where $D_{pp} = -1$; it matters, since $D_{pp} = -1$ does occur (e.g. at $t = 0.05$).
- The cost of one UCCSD energy evaluation for $L = 6$ on Exaqt is about 0.6 s on one core; a converged run needs 10^3–10^4 evaluations.